# ZOO405 – ECOLOGY 100% GUESS

## Final-Term File 2023 By

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## Igneous and sedimentary rocks?

**Igneous rocks** are formed by the cooling of hot molten rock also known as magma. When the hot magma cools it begins to harden meaning once it had fully cooled it create what is known to be an igneous rock.

**Sedimentary rocks** are formed from pre-existing rocks. When rocks erode and mix with other dirt, clay and particles then settle together the mix together to form a sedimentary rock.

## Characteristics of Population Ecology:

- Population ecology is the study of populations in relation to the environment. Includes environmental influences on population density and distribution, age structure, sex relation, and variations in population size.
- All species occur in groups called populations. There are certain characteristics that all populations have, such as population density, spatial distribution, and growth rate. These characteristics are used to classify all populations of organisms, including bacteria, animals, and plants..
- Population density one characteristic of a population is its population density, which is the number of organisms per unit area.
- Another characteristic of a population is called dispersion—the pattern of spacing of a population within an area.

## 3 sources of noise pollution:

Industrialisation, vehicles, events, construction sites, loudest speakers.

## What is the ozone layer and ozone depleting substances:

**The ozone layer** is the common term for the high concentration of ozone that is found in the stratosphere around 15–30km above the earth's surface. It covers the entire planet and protects life on earth by absorbing harmful ultraviolet-B (UV-B) radiation from the sun.

**The main reasons** for the ozone hole are chlorofluorocarbons, carbon tetrachloride, methyl bromide and Halocarbon refrigerants , Solvents , Propellants in aerosols, Hydro chlorofluorocarbons, Methyl bromide, Carbon tetrachloride, Methyl chloroform.

## Ecotone and its importance?

An ecotone is an area that acts as a boundary or a transition between two ecosystems. A common example could be an area of marshland between a river and its riverbank.

## Importance:

- Ecotones are of great environmental importance. Because the area is a transition between two ecosystems or biomes, it is natural that it contains a large variety of species of fauna and flora as the area is influenced by both the bordering ecosystems.
- They can act as buffer zones offering protection to the bordering ecosystems from possible damage. For example, a wetland can absorb pollutants and prevent them from seeping into the river.
- Ecotones are also a sensitive indicator of global climate change.
- They have a greater variety of organisms.
- They also offer a good nesting place for animals coming in search of a nesting place or food.
- They serve as a bridge of gene flow from one population to another because of the larger genetic diversity present.

## Explain water pollution:

Water pollution happens when chemicals or dangerous foreign substances are introduced to water, including chemicals, sewage, pesticides and fertilizers from agricultural runoff, or metals like lead or mercury. According to National Oceanic and Atmospheric Administration (NOAA), 80% of pollution in marine environment comes from the land through sources like runoff. Water pollution can also severely affect marine life. For example, sewage causes pathogens to grow, while organic and inorganic compounds in water can change the composition of the precious resource. More than 50 kinds of diseases are caused by poor drinking water quality, and 80% of diseases and 50% of child deaths are related to poor drinking water quality in the world. However, water pollution causes diarrhea, skin diseases, malnutrition, and even cancer and other diseases related to water pollution.

## What is Effect of noise pollution on human health?

Noise pollution can be hazardous to human health in the following ways:

- Hypertension: It is a direct result of noise pollution which is caused due to elevated blood levels for a longer duration.
- Hearing loss: Constant exposure of human ears to loud noise that are beyond the range of sound that human ears can withstand damages the eardrums, resulting in loss of hearing.
- Sleeping disorders: Lack of sleep might result in fatigue and low energy level throughout the day affecting everyday activities. Noise pollution hampers the sleep cycles leading to irritation and an uncomfortable state of mind. Page 218 of 295.
- Cardiovascular issues: Heart-related problems such as blood pressure level, stress and cardiovascular diseases might come up in a normal person and a person suffering from any of these diseases might feel a sudden shoot up in the level.

## Sources of noise pollution long:

Industrialisation, vehicles, events, construction sites, loudest speakers, some other appliances also contribute to noise pollution, like television, transistor, radio, etc.

## Difference between random and uniform dispersion?

**Uniform dispersion** refers to organisms that live in a specific territory. One example would be penguins. Penguins live in territories and within those territories the birds' space themselves out relatively uniformly.

**Random dispersion** refers to the spread of individuals such as wind-dispersed seeds, which fall randomly after traveling.

## How rangeland distinguish from pastureland:

**Rangelands** are distinguished from pastureland by the presence on them of native vegetation, rather than of plants established by human societies, and by their management principally through the control of the number of animals grazing on them, as opposed to the more intensive agricultural practices of seeding, irrigation, and the use of fertilizers.

## Steps of sulphur cycle:

The sulfur cycle is made up of 4 steps: mineralization, oxidation, reduction and incorporation.

Workingly

1= Decomposition of organic compounds 2= Oxidation of hydrogen sulphide to elemental sulphur 3= Oxidation of elemental sulphur 4= Reduction of sulphates.

## How to prevent water pollution?

Simple small changes made domestically will make a huge difference. Treatment of sewage: Treating waste products before disposing of it in water bodies helps reduce water pollution on a large scale. Agriculture or other industries can reuse this wastewater by reducing its toxic contents.

1= Turn off the tap when running water is not necessary 2= Use environmentally household products 3= Avoid over dosage of pesticides and fertilizers 4= Don't throw the non-degradable substances to rivers, lakes or oceans 5= Pretreatment of industrial waste water before leaving to water bodies.

## Objective of Pakistan environmental act 1997:

**The PEPA' 1997** provides the framework for implementation of NCS, establishment of Provincial Sustainable development Funds, Protection and conservation of species, conservation of renewable resources, establishment of Environmental Tribunals and appointment of Environmental Magistrates, Initial Environmental Examination (IEE), and Environmental Impact Assessment (EIA).

## Explain use of desalination water in agriculture?

- In spite of this development, the costs of desalinated water are still too high for the full use of this resource in irrigated agriculture, with the exception of intensive horticulture for high-value cash crops, such as vegetables and flowers (mainly in greenhouses), grown in coastal areas (where safe disposal is easier than in inland areas).
- For agricultural uses, RO is the preferred desalination technology because of the cost reductions driven by improvements in membranes in recent years. Spain provides a significant example of the application of desalinated water in irrigation.
- Most of these plants process brackish water (only 10 percent of the total desalinated water for agriculture originates from seawater).
- In this country, small and medium-sized brackish water desalination plants, with a capacity of less than 1 000 m3 /d (11.6 litres/ s), are common because they adapt better to individual farmer requirements and to the existing hydraulic structures.

## Difference between abiotic and biotic factors?

**Abiotic components** are non-living chemical and physical factors of an ecosystem. A few examples of abiotic components include sunlight, soil, air, moisture minerals.

**Biotic components** are living factors of an ecosystem. A few examples of biotic components include bacteria, animals, birds, fungi, plants, etc.

## Difference between predator and prey with example?

**Predator:** A predator is an organism that consumes all or part of the body of another living or recently killed organism, which is its prey. **Prey:** the organism that is eaten by the predator is called prey. Example: If we see a lion eating a zebra, we can say that the lion is a predator, and zebra is prey.

## 1st law of thermodynamics:

**The First Law of Thermodynamics**, also known as the law of conservation of energy, states that energy can neither be created nor destroyed. It may change from one form to another, but the energy in a closed system remains constant.

**Ozone Depletion:** When chlorine and bromine atoms come into contact with ozone in the stratosphere, they destroy ozone molecules. One chlorine atom can destroy over 100,000 ozone molecules before it is removed from the stratosphere. Ozone can be destroyed more quickly than it is naturally created.

## Coniferous and deciduous forests?

**Deciduous trees** have broad leaves that change color in the fall and spread their seeds using flowers. These forests are found in America, Western and Central Europe, and Northeastern Asia where the winters are cold and summers are warm.

**Coniferous trees** have needles instead of leaves, they don't change color in the fall, and they use cones instead of flowers to spread their seeds (example: pine, fir, and hemlock). These are found mostly in the northern parts of North America, Asia, and Europe.

## Define community population and ecosystem population and how they different from & each other explain?

**Population Ecology:** It deals with factors that alter and impact the genetic composition and the size of the population of organisms. Ecologists are interested in fluctuations in the size of a population, the growth of a population and any other interactions with the population.

**Community Ecology:** It deals with how community structure is modified by interactions among living organisms. Ecology community is made up of two or more populations of different species living in a particular geographic area.

## Impacts of air pollution on health?

The air we breathe has a very exact chemical composition; 99% of it is made up of nitrogen, oxygen, water vapor and inert gases. Air pollution occurs when things that aren't normally there are added to the air. A common type of air pollution happens when people release particles into the air from burning fuels. This pollution looks like soot, containing millions of tiny particles, floating in the air. Long-term health effects from air pollution include heart disease, lung cancer, and respiratory diseases such as emphysema. Air pollution can also cause long-term damage to people's nerves, brain, kidneys, liver, and other organs. Some scientists suspect air pollutants cause birth defects.

## Prevention for noise pollution?

1= Some noise pollution preventive measures are provided in the points below. 2= Honking in public places like teaching institutes, hospitals, etc. should be banned. 3= In commercial, hospital, and industrial buildings, adequate soundproof systems should be installed. 4= Musical instruments' sound should be controlled to desirable limits. 5= Dense tree cover is useful in noise pollution prevention. 6= Explosives should not be used in forest, mountainous and mining areas.

## Difference  primary/secondary pollutants?

The pollutants that directly cause air pollution are known as primary pollutants. **Examples:** Nitrogen oxides (NOx), Carbon monoxide (CO), Volatile organic compounds (VOCs), Sulfur oxides (SOx), Particulate matter (PM), Mercury.

The pollutants formed by the intermingling and reaction of primary pollutants are known as **secondary pollutants**. Smog, formed by the intermingling of smoke and fog, is a secondary pollutant. Examples of a secondary pollutant include ozone, which is formed when hydrocarbons (HC) and nitrogen oxides (NOx) combine in the presence of sunlight; NO2, which is formed as NO combines with oxygen in the air; and acid rain, which is formed when sulfur dioxide or nitrogen oxides react with water.

## Global ecology:

It deals with interactions among earth's ecosystems, land, atmosphere and oceans. It helps to understand the large-scale interactions and their influence on the planet.

## Profundal Zone:

This zone extends beyond the Limnetic zone and is a deeper region of the water. Here, there is usually very little or no sunlight. It is cooler. Living things in this region depend on drifting organic matter for food.

## Desertification:

**Desertification,** also called desertization, the process by which natural or human causes reduce the biological productivity of drylands (arid and semiarid lands).

**Cause:** Deforestation, Farming, Excessive use of fertilizers and, pesticides, Animal grazing, Global warming, Overpopulation, Changes in land use, Mining, Excessive consumption, Waste production and disposal, Soil pollution, Acid rain, Natural causes.

**Solution:** Stricter regulations regarding farming practices, Reduction in fertilizer and pesticides, Reforestation, Population control, Technological advancements Confinement of mining practices, Avoidance of pollution, Adjustments in consumption, Avoid waste production, Education, Convince others

| Ecosystem | Community | Population |
|---|---|---|
| An ecosystem is a geographic area where plants, animals, and other organisms, as well as weather and landscape, work together to form a bubble of life. | A community is a group of different species living in the same area. | Population is a group of the same species living in the same area. |
| Ecosystems contain biotic or living, parts, as well as abiotic factors, or nonliving parts. | Communities may share a sense of place situated in a given geographical area | It can determined by such factors as density, sex ratio, birth and death rates, emigration, and immigration. |

## Economical Importance of Grasslands:

1= Grasslands are the grazing areas of many rural communities. 2= Farmers who keep cattle or goats, as well as shepherds who keep sheep, are highly dependent on grasslands. 3= The grass is also used to thatch houses and farm sheds.4= Overgrazing by huge herds of domestic livestock has degraded many grasslands. 5= Grasslands have diverse species of insects that pollinate crops.

## What is evaporation:

Evaporation is the process that changes liquid water to gaseous water (water vapor). Water moves from the Earth's surface to the atmosphere via evaporation. Evaporation occurs when energy (heat) forces the bonds that hold water molecules together to break.

## What do you mean by precipitation?

Precipitation is rain, snow, sleet, or hail — any kind of weather condition where something's falling from the sky. Precipitation has to do with things falling down, and not just from the sky. It's also what happens in chemical reactions when a solid settles to the bottom of a solution.

## Which three main strategies need ecologists:/ approaches to study ecological data.

Bservation, experimentation, and modeling: Observation often leads to questions and hypotheses. Experiments can be used to test hypotheses. Modeling helps ecologists understand complex processes.

## Nitrogen fixation:

It is the initial step of the nitrogen cycle. Here, Atmospheric nitrogen (N2) which is primarily available in an inert form, is converted into the usable form -ammonia (NH3).

## Objectives of environmental:

Protect and conserve critical environmental resources and invaluable natural and man- made heritage which are essential for life-supporting livelihoods and welfare of the society.

## Resources of competition:

Food, water, shelter, light, territory, and substrate.

## Characteristics of population:

Population characteristic commonly include age, gender, race, and ethnicity. Population characteristics help predict the possible health outcomes and risk for certain diseases. For example, older populations are more at risk for strokes and heart attacks.